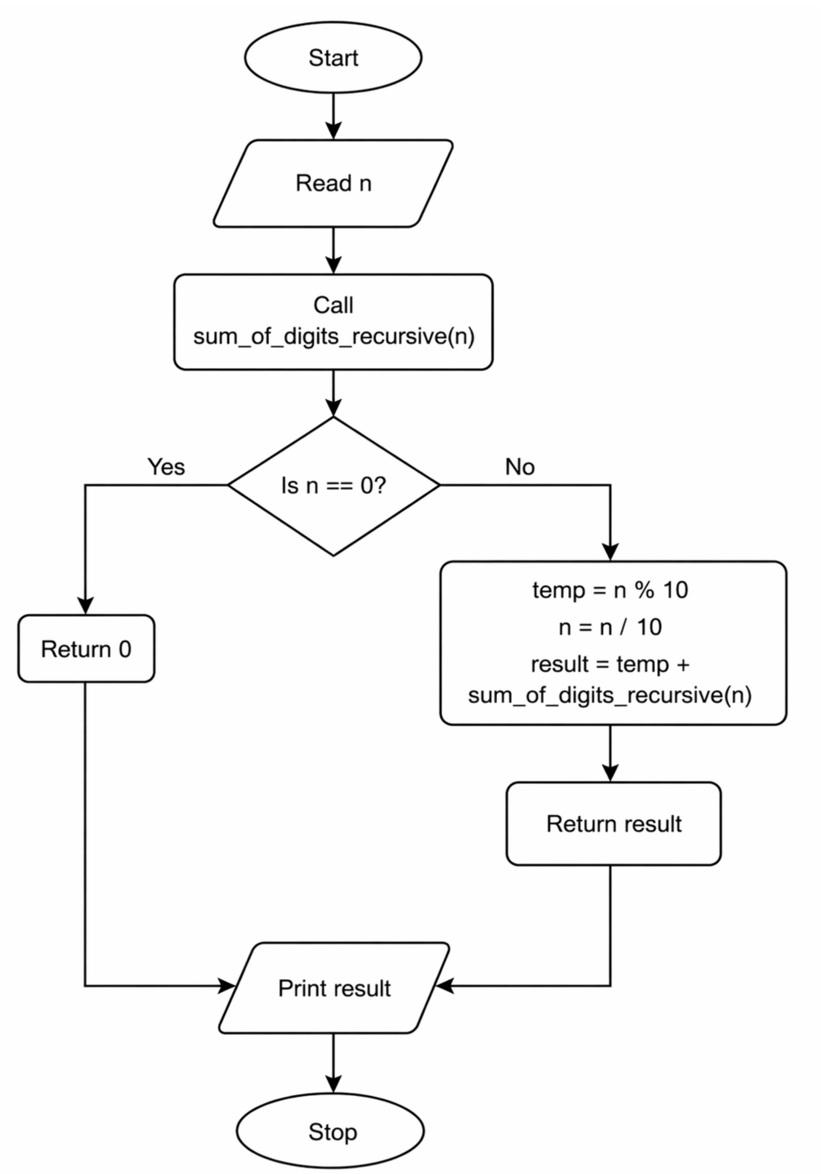
11.1.2. Sum of Digits using Recursion

AIM:Write a recursive function sum_of_digits_recursive(n) that calculates the sum of the digits of a given non-negative integer n.

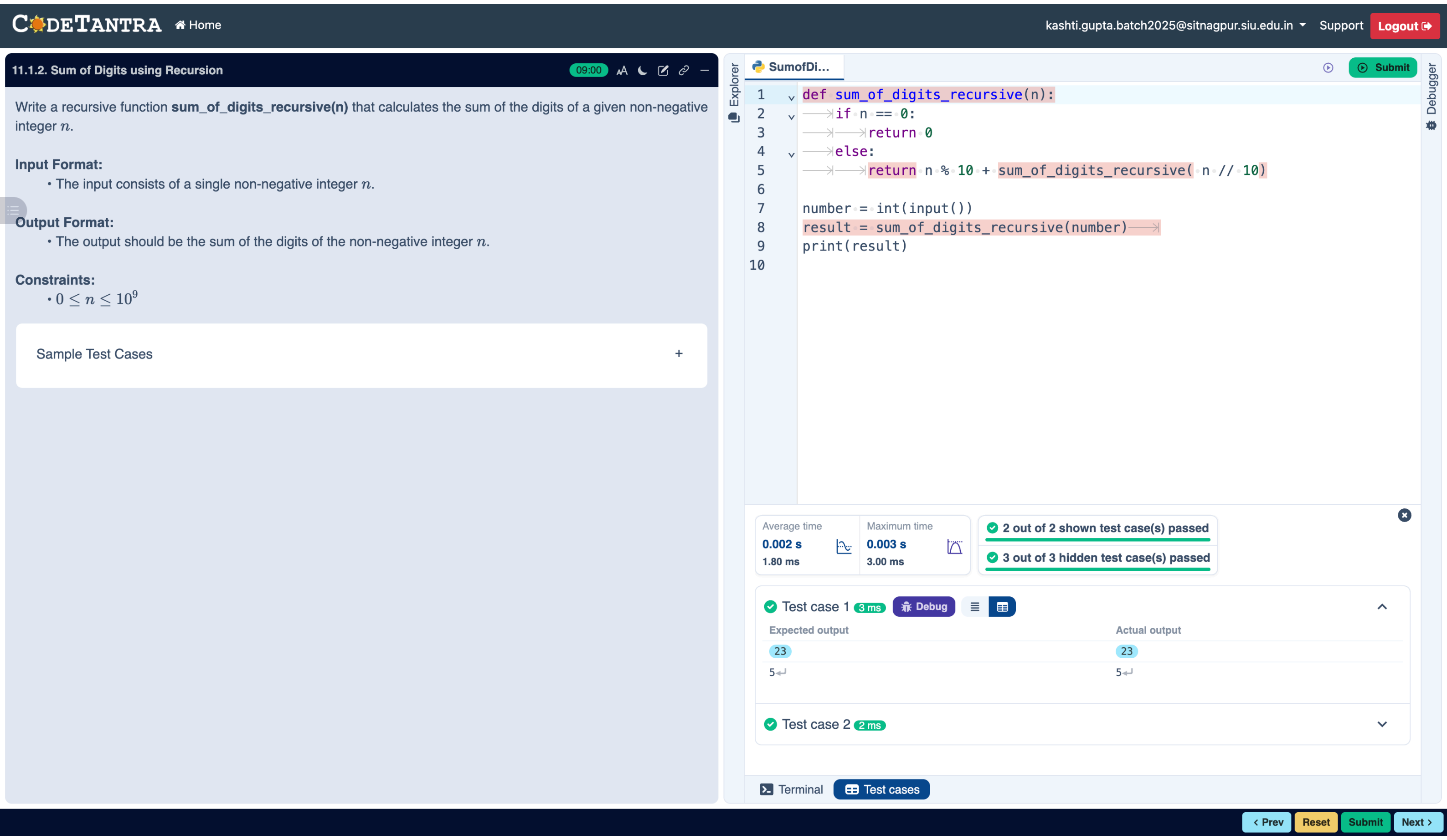
Algorithm

1.Start
2.Read number n
3.If n == 0 → return 0
4.Else → return last digit + recursive call
5.Call function
6.Print result
7.Stop

FLOWCHART



EXECUTION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11.1.2. Sum of Digits using Recursion
09:00
Write a recursive function sum_of_digits_recursive(n) that calculates the sum of the digits of a given non-negative integer n.
Input Format:
• The input consists of a single non-negative integer n.
Output Format:
• The output should be the sum of the digits of the non-negative integer n.
Constraints:
• 0 ≤ n ≤ 10^9
Sample Test Cases
SumofDi...
Submit
def sum_of_digits_recursive(n):
if n == 0:
return 0
else:
return n % 10 + sum_of_digits_recursive(n // 10)
number = int(input())
result = sum_of_digits_recursive(number)
print(result)
Average time
0.002 s
1.80 ms
Maximum time
0.003 s
3.00 ms
2 out of 2 shown test case(s) passed
3 out of 3 hidden test case(s) passed
Test case 1
3 ms
Debug
Expected output
Actual output
23
5
23
5
Test case 2
2 ms
Terminal
Test cases
Prev
Reset
Submit
Next